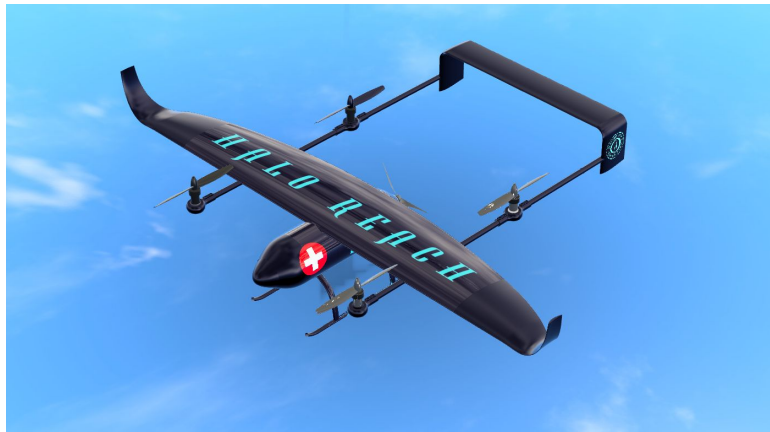

HALO REACH: Design and Optimisation of VTOL Rotor with Computational Analysis

UAS for Disaster Relief - AE3 2024



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Wing Aerofoil Selection	Max	Max
Main Wing Design	Matthew and Max	Matthew
Winglet Design	Naboth	Naboth
Adkins and Leibeck Code	Parasdeep, Naboth, Shahriar	Parasdeep
Propellor Initial Thrust/Power analysis	Parasdeep	Parasdeep
Propeller Design	Parasdeep	Parasdeep
Propeller performance analysis	Parasdeep	Parasdeep
Rotor Initial Thrust/Power analysis	Shahriar	Shahriar
Rotor Design	Shahriar	Shahriar
Rotor Performance Analysis	Shahriar	Shahriar
CFD Setup and Validation For Stock Rotor	Matthew, Max	Matthew
Rotor CFD	Shahriar, Matthew	Shahriar
Propeller CFD	Matthew	Matthew
Full Wing CFD	Naboth	Naboth
Full Aircraft CFD	Max	Max
Blanketing Affects of Tailplane	Naboth	Naboth
Tailplane CFD	Naboth	Naboth

Table 2. Separation of Work in Aero Team

Abstract

This report aims to outline the methods used for the first iteration design of the VTOL rotors and analyse the performance utilising both computational and analytical means. Modern Blade Element Momentum Theory was used to produce the optimum chord and twist distribution for the parameters based on parametric studies and mission requirements. The blade performance was then analysed via BEM methods and Computational Fluid Dynamic simulations for thrust and power requirements. The findings and conclusions in this report provide valuable recommendations for the design and confirm that the rotor was successfully designed and could meet performance requirements.

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1 Introduction

1.1 Mission overview

Following natural disasters, conventional transport routes may become inaccessible and can no longer be used as an option to provide disaster relief. Historically, these situations relied on aerial support from helicopters but the landscape is changing with small drones beginning to be used. Small scale UAV's are even being proposed for more casual use within urban environments, with EASA proposing plans for 'vertiport' designs to facilitate and easily integrate the landing of such vehicles in city centres and deliver assistance when required [1].

This report is part of a project to develop a versatile, modular Vertical Take-Off and Landing (VTOL) Uncrewed Aerial System (UAS) that can conduct:

- Vertical take-off and landing from ship decks even under high-wind and gusty conditions.
- Cruise to and from the site of a disaster.
- Serve as a long-endurance communications relay, OR land vertically to deliver relief supplies.

This report specifically focusses on satisfying the VTOL requirement, and is a blueprint for a rapid design procedure and analysis, combining both analytical and computational techniques for a holistic methodology. There were certain requirements for the mission that were especially important in the aerodynamics design process to inform sizing constraints shown in Table 3.

Mission requirement	Value
GTOW	160kg
Launch and Recovery Footprint	6x6 m

Table 3. Specifications

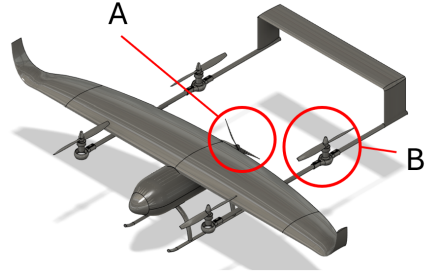


Figure 3. Final Design

Figure 3 shows the final configuration after the conceptual design and object B is referred to as the VTOL rotors which will be the focus of this report. The initial VTOL rotor design was done individually and was set to a deadline of 10 working days from the point of idea conception. The pusher propeller was given an extra few days for design which allowed for more parameter studies to be conducted. Most constraints on sizing came from collaboration with other teams especially structures and flight dynamics, so this was not necessary for the VTOL design.

1.2 Objectives

This report will be a holistic discussion on the VTOL rotor design. Firstly, a literature review was conducted to discover existing rotor design methodology and gain an brief overview of key parametric studies. Secondly, an airfoil optimization study was conducted to select suitable inboard and outboard aerofoils and final optimization parameters were selected as input into the optimization algorithm to create an initial blade geometry which was refined using QBlade. The performance of the blade was then analysed using both analytical and computational methods. The success of the design was established by meeting the desired thrust level at the target velocity, whilst ensuring the power requirement could be met with the selected motor.

1.3 Literature review

1.3.1 Blade Element Momentum Theory

Blade Element Momentum (BEM) Theory was first formulated in a useful form by Glauert [2]- it combines the Actuator Disc Theory (Momentum Theory) and Blade Element Theory. Under BEM Theory, certain assumptions for the inflow must be applied such as being steady, incompressible, assymmetric and inviscid. Further details on the theory can be found in Appendix A. In this project, BEM was mainly used to predict performance but the optimization of the blades followed a more modern method, which combines BEM theory and also implements the Betz condition for minimum energy loss, to produce the most efficient propeller/rotor design. More details on this are in the next section.

1.3.2 Optimum Rotor Design

The methodology in 'Design of Optimum Propellers' can be used to find the optimal chord and twist distributions with a set of input parameters that are shown in Table 4.

Input optimization parameters	Output
Aerofoil properties	Chord distribution Twist distribution
RPM	
Hub radius	
Tip radius	
Tip Velocity	

Table 4. List of Inputs and Outputs for the Adkins algorithm

This modern approach in [3] is especially relevant for the VTOL rotors which are likely to encounter high induced velocities relative to the freestream and thus would fail the light loading assumptions of previous classical theory. This method also removes the need for small angle approximation assumptions. An iterative method is required to calculate vortex displacement velocity and flow angle distribution. A summary of the implementation steps are found in [4]. For a more accurate model, radial flow momentum losses can be estimated using either the Prandtl or Goldstein momentum loss functions. In our case, the Prandtl tip loss function was used, seen in Equation 1:

$$F = \frac{2}{\pi} \arccos(e^{-f}) \quad (1)$$

where

$$f = \frac{B(1 - \zeta)}{2 \sin(\phi_t)} \quad (2)$$

The Prandtl tip loss factor aids in modelling the helical vortex wake pattern as vortex sheets that are convected by the flow but do not have a direct impact on the wake itself [5]. This is an important correction to the turbine wake model that brings the BEM approximations closer to experimental data. In the Appendix, the potential of a hub correction factor is also considered and the validity of BEM assumptions and underlying physics will be discussed again.

2 VTOL Rotor Design and Initial Trade Studies

2.1 Design Goals

This drone requires the ability to hover during drop offs or rescues. Based on discussions with the Propulsion Team and their preliminary research regarding available motors and also mission requirements, the target velocity was set to 1m/s. A suitable acceleration of 0.5 m/s² was chosen by the Propulsion team, which led to a thrust target of 412N/rotor for the first design iteration (T/W=1.05).

2.2 Initial Studies and Design procedure

An extensive literature review was conducted to investigate typical rotorcraft design parameters. Initial calculations and empirical data were used to set parameters which could be later refined, such as Reynolds number distributions, design lift coefficients and other blade parameters (i.e number of blades). The rotor design required high interoperability and good communication to accommodate the requirements of multiple teams, especially Propulsion, Flight Dynamics and Structures. The following list describes the steps used to produce the rotor design:

1. Estimate Reynolds and Mach distributions along the blade.
2. Select optimum aerofoil for these distributions.
3. Finalize RPM, number of blades, and sizing for optimization code based on parametric studies and requirements from other sub teams.
4. Utilise in-house optimization methodology (in appendix) to find the optimum chord and twist distributions, then further adjust design and rapidly test using QBlade to ensure thrust and power within requirements.
5. If performance not adequate, return to step 3 and adjust optimization parameters.

2.3 Initial Parametric Studies

This is a general configuration study used in the design process to help inform parameter choices to minimise power requirements. The values for power here are not representative of the final design but are used to directly compare the effect of different parameters. Further parametric studies were not required at this stage.

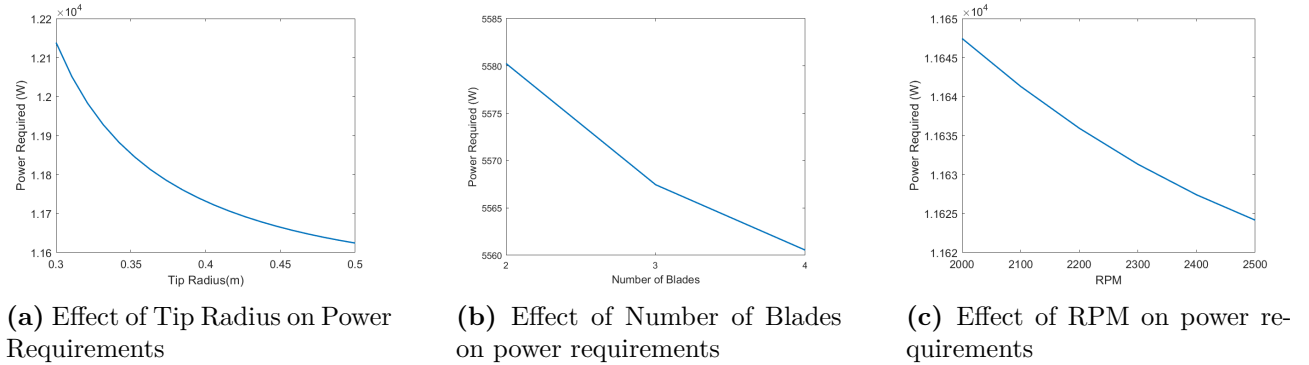
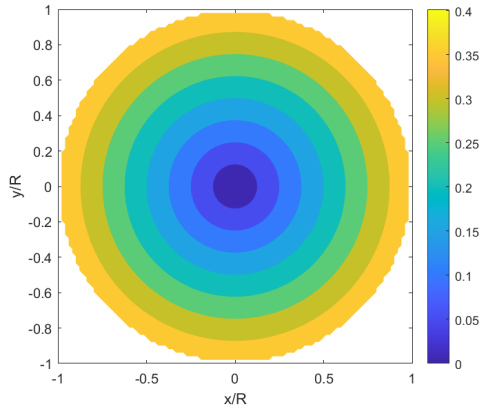


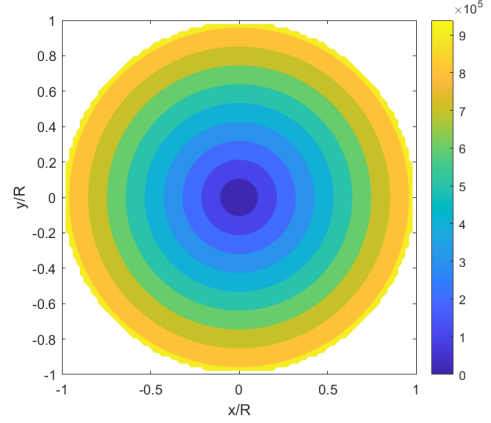
Figure 4. Initial Parametric Studies

2.4 Airfoil Selection Methodology

Airfoil selection began with initial analyse of Mach number (M) and Reynolds (Re) variation across the rotor disk. For this type of configuration, typically low Re refer to regimes $< 10^6$ and the majority of the blade experienced this as seen in Figure 5. This would better inform our airfoil choice using XFOIL data. For a $M > 0.3$, compressibility effects would have to be considered. A combination of both qualitative and quantitative factors affected the preliminary choice of RPM. For example, it was important to ensure that there was a sufficient margin between the drag divergence mach number for the chosen airfoils and M_{tip} while also considering that excessive RPM may also create structural issues later in the design process.



(a) Mach variation across rotor disk



(b) RE variation across rotor disk

Figure 5. Initial Analysis of Mach and Re number distributions

2.4.1 Aerofoil selection criteria

Early into the design process, decisions were made for some constraints of this UAV such as a maximum tip radius of 0.5m and minimizing rpm as both a precautionary aerodynamic drag and structural measure. Furthermore, as is common in rotary design, it was decided to choose an inboard airfoil and then an outboard airfoil due to different design requirements along the blade.

For inboard aerofoils

- good performance at low Re , low M such as high $(L/D)_{\max}$, $C_{L_{\max}}$
- $t/c > 10\%$ to absorb high structural loads at the root

For outboard aerofoils

- good performance at high Re , high M such as high $(L/D)_{\max}$, $C_{L_{\max}}$
- thickness not initially considered
- good stall characteristics
- low to medium camber ($< 4\%$)

The effect of camber on the outboard airfoil could be a cause for excessive blade twisting and would require high control loads so the outboard aerofoil camber was set to be a maximum of 4% [6]. Thickness was also an important criteria but specifically for the inboard airfoil and aerofoils below a t/c of 10% were disregarded. The airfoils considered are typical helicopter, experimental and turbine airfoils. The significant portion of the blade lift would be produced in the outer part of blade; this section would thus operate at higher angles of attack meaning good stall characteristics were desirable.

Table 5 showcases some properties for shortlisted airfoils. After comparing these initial parameters, a further parametric study was used to compare between aerofoils using an quick estimate for the Figure of Merit (FM). In low Reynolds number environments, it is important to consider losses that impact hover performance such as: induced and profile drag losses, nonuniform flow, slipstream rotation and tip losses [6].

$$FM = \frac{C_t^{3/2}}{C_p \sqrt{2}} \quad (3)$$

FM is defined in Equation 3 where C_t and C_p are thrust and power coefficients respectively. It is a measure of efficiency for a rotor generating thrust at a give power and is the ratio of ideal hover power over actual hover power [6]. The Figure of Merit was generated between some aerofoils of particular interest but further qualitative factors were also considered and will be discussed in the next section. The FM seen here is not representative of the final design and it is only used as a comparative measure.

Aerofoil	t/c (%)	Cl_{max}	$(L/D)_{max}$	Stall angle (°)
SG6043	10	1.7	143	15.75
Clark-Y	11.7	1.43	98.7	14.25
GOE 804	6	1.62	169.1	11
GOE 448	12.8	1.8	119.3	10.5
e58	5.6	1.6	175.1	-
VR-7	12	1.3	103.5	15
RC-10	10	1.17	84.3	13
VR-8	8.2	0.95	56.2	9.75
NACA 23012	12	1.43	77.2	14.5
NACA 0012	11	1.22	80.7	16
E216	10.4	1.65	136.84	14.5
AH7476	5.9	1.65	148.4	13
BOEING 23010	10.2	1.33	79.9	14

Table 5. Aerofoil Characteristics

Aerofoil	Figure of Merit
GOE 804	0.837
VR-8	0.81
NACA 23012	0.77
AH7476	0.84
BOEING 23010	0.76

Table 6. Outboard Aerofoil Figure of Merit

Aerofoil	Figure of Merit
Clark Y	0.83
GOE 448	0.8
VR-7	0.79

Table 7. Inboard Aerofoil Figure of Merit

2.5 Final Choice of aerofoil

The considered inboard aerofoils are the Clark Y, GOE 448 and VR-7. The GOE 448 had very good aerodynamic properties and high thickness, but is an experimental aerofoil so was disregarded. The Clark-Y has been used on aircraft and also gliders due to high Cl_{max} . The VR-7 was specifically designed for rotorcraft but has a much lower Cl_{max} with lower $(L/D)_{max}$. In conclusion, the Clark-Y was chosen for the inboard aerofoil over the VR-7 due to superior aerodynamic properties and slightly better Figure of Merit.

For the outboard aerofoil choice, the GOE 804 had very good aerodynamic properties but was disregarded for two reasons. Firstly, once again it is an experimental aerofoil with virtually no use within industry and secondly because it had very excessive camber of 6.3%. The outboard aerofoil is most important for lifting capabilities of the blade so the VR-8 was disregarded eventually due to poor aerodynamic properties seen on Table 5. Although, the AH7476 aerofoil had the best hover performance as seen in Table 6 and very good aerodynamic properties, there was a lack of accessible experimental data for the aerofoil. Although thickness was not too significant for the outboard aerofoil, the structures team at this point expressed concerns and pushed for a thicker design. Thus the NACA 23012 was preferred as it had fairly comparable Cl_{max} . Experimental data also suggested low pitching moments when using this aerofoil. Upon further research, the BOEING 23010 was considered, but this lacked publicly available test data so the NACA 23012 was the final choice for the outboard aerofoil. This aerofoil design also has a higher thickness so would be more convenient for the structures team.

2.6 Final Selection of Optimization Parameters

The table below shows the final choice of optimization parameters. The rest of the subsection is a brief explanation on the rationale behind the selection.

Design Parameters	Value
Thrust required (N /rotor)	412.00
Optimized for RPM	2500.00
Tip radius (m)	0.50
Hub radius (m)	0.08
Tip Velocity (m/s)	130
Number of blades (m/s)	2

Table 8. Design Parameters

2.6.1 RPM

After selecting the airfoils, we could now get a better estimate of the required RPM. The average C_l along the blade was found as six times the blade loading where the blade loading was 0.13 [6]. It was important to ensure that there was a sufficient margin between the drag divergence mach number, M_{dd} , for the chosen aerofoils ($=M_{0.64}$) and M_{tip} , such that we avoided unnecessary drag. It was also important to consider that excessive RPM may also create structural issues.

At takeoff weight, for hover, and knowing that the majority of blade lift is produced towards the tip of the blade, an estimate for the tip velocity and thus rotational speed could be found using basic lifting equations. This provided an approximate RPM of 2100 to generate the required thrust for hover and provided an estimate of the tip velocity. Based on the RPM study in Figure 4c, we see how optimizing for a higher RPM reduces the power requirements based on methodology in [3]. So an optimized RPM of 2500 was selected. Optimizing for a higher RPM > 2500 gave a chord geometry that was excessively small and below manufacturing capabilities. Power savings below this point were negligible. It is important to note that this RPM is used as an input to find the optimum chord and twist distributions. The final blade will be a variable RPM design such that it can operate at alternative thrust requirements for different maneuvers.

2.6.2 Number of Blades and Sizing

For the VTOL rotors, it was preferred to use a 2-blade design for the following reasons:

- During cruise the VTOL rotors would be idle, hence 2 blades could be aligned into a streamlined position to minimise drag. More blades would also increase complexity of hub design.
- It was a more optimum design since although increasing blade number would reduce power requirements (Figure 4b), the potential mass savings by selecting a lighter motor could not justify the mass increase from an additional blade.

For blade sizing, in order to minimize power requirements for a given RPM and target thrust, Figure 4a suggests to maximise tip radius. The flight dynamics team imposed a tip radius constraint of 0.5m based on optimum rotor placement on the wings from a stability perspective, so this maximum value was chosen. The hub radius was determined from empirical data and set to a value of 0.08m [6].

3 Final Rotor Geometry and Performance

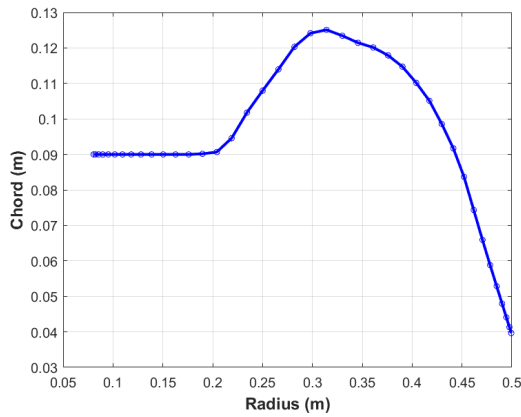


Figure 6. Final Design

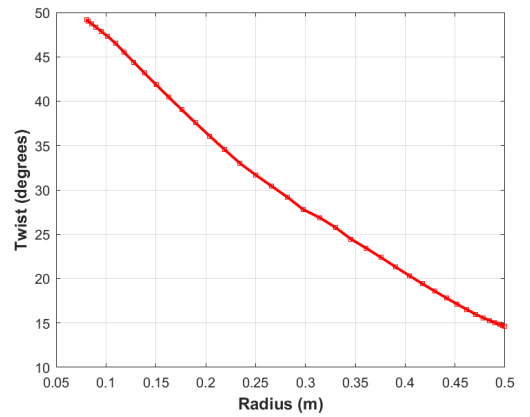
In order to attain the final geometry, in-house optimization code was used as a starting point and then the geometry was further optimised on Qblade since it could better model the use of multiple aerofoils, and also model the interpolated region between the inboard and outboard airfoils. The GUI allowed us to quickly iterate through slight adjustments in the design and receive immediate results. QBlade had been validated against [3] as it followed the same optimization method. The aerofoil variation is as follows:

- Clark-Y for the inner 40% of the blade.
- 10% interpolation region between NACA 23012 and Clark-Y with 0.5 interpolation factor
- NACA 23012 used for the remaining 50% of the blade

As mentioned previously, the timeline for the VTOL rotor was necessary to allow other teams to commence their calculations promptly, thereby removing the opportunity for further iterations. Consequently, this made it important to anticipate potential design issues in advance, particularly those related to structural integrity.



(a) Final Chord Distribution



(b) Final Twist Distribution

Figure 7. Final Geometry

Figure 7 showcases the final geometry. Figure 7a shows the chord distribution where the first 10% of the chord was manually set thicker to 0.09m. This would help prevent stress concentrations building up which would damage the longevity of the rotors. The idea is that in future iterations the blade could be made thinner closer to the root which is more representative of the ideal optimization in [3]. Furthermore, the ideal twist was also exceptionally high initially, starting at close to 85° at the root, so a twist offset of -36° was used along the blade, for ease of manufacturing. Excessive blade twisting would also affect negatively impact the controllability of the rotors. Once the design was established, the Structures Team added a hub design and the final CAD is seen in Figure 6.

3.1 BEM performance predictions

The key criteria to measure the success of the design was whether it could meet the thrust requirement of 412N. During the design process, care was taken to choose parameters that would also reduce power requirements of the blade. Both QBlade, XROTOR and an in-house BEM code were used to validate thrust estimates. Estimates were relatively close, with slight differences (see appendix B). For example, Xrotor did not allow the input of multiple aerofoil properties along the blade; it could also not accurately simulate the interpolated geometry and underestimated the lifting characteristics of the blade overall. QBlade software was most accurate, as it uses the correct lift and drag polars at the correct Reynolds number for each section of the blade and could generate interpolated airfoil properties. QBlade also provided a better estimate than the in-house code since this required an interpolation of aerodynamic parameters and creates invalid predictions at low velocities. The final performance results (using QBlade) for BEM are shown in Table 9 and the final blade had an FM=0.61.

Thrust (N)	Max Power Required (kW)
449	10.2

Table 9. Final BEM performance

Figure 8 provides additional data, for the thrust the blade can generate at a fixed RPM of 2500 if the desired climb velocity was adjusted and Figure 9 showcases a thrust available at different RPM's.

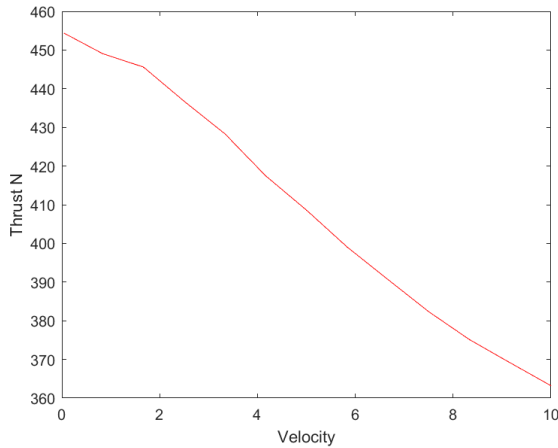


Figure 8. Thrust available at different climb velocities at fixed RPM

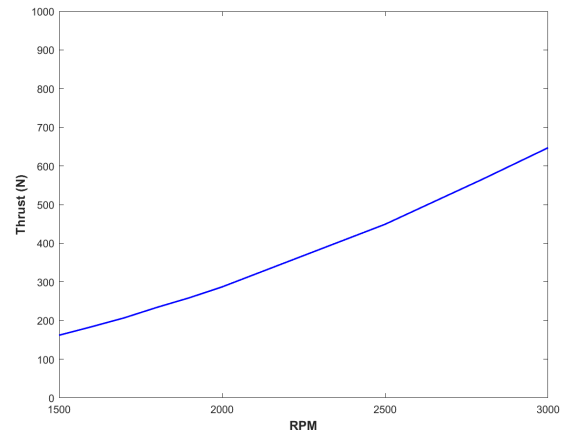


Figure 9. Thrust available at varying RPM at fixed climb velocity

4 Computational Fluid Dynamics

BEM methods are computationally fast and easily programmable. Over time, advanced CFD methods evolved based on the Navier-Stokes equations. HPC resources allowed the complexity of CFD methods

to increase, by utilising parallel processing to reduce computation time and was especially useful in this project, although licensing access still constrained simulations. Star-CCM software was used to perform CFD simulations in this report. For validation, CFD results for a stock rotor were compared against wind tunnel data and more details can be found in [7].

4.1 CFD setup

4.1.1 Geometry and Mesh Design

The geometry of the rotor is seen in Figure 6. The setup utilised a bullet domain, with a large cylindrical fluid domain with a rotating rotor domain within it. An outer region 'fluid mesh' was created to simulate the fluid flow in the atmosphere. An inner region 'rotating mesh' was created and confined to the actual rotor.

For the rotating mesh, the surface remesher helped to create a conformal mesh. A polyhedral mesher was chosen as a good measure of speed, efficiency and quality [8]. It helped to model the curved surfaces of the rotor and prismatic layer cells were used next to the boundary surface to accurately capture behaviour within the boundary layer. For the 'fluid mesh', a trimmed cell mesher was used as this is useful in creating flow aligned meshes along with surface remesher [8]. More details on the geometry and mesh can be found in Appendix D.

4.1.2 Models

The selection of appropriate physics models is crucial for producing useful results that accurately simulate the environment that the VTOL rotor will operate in, while also removing the simplifications that BEM makes. Some of the key models will be discussed here but further details can be found in [7]. Ideal Gas model is chosen since the mach number towards the tip of the blade exceeds $M 0.3$; this model can accurately simulate the compressibility effects experienced due to the variations in density, temperature and pressure in the rotor domain. Hence a coupled solver is also used since the interdependencies between momentum and energy become more significant in compressible flow. Implicit unsteady model is used since the flow properties change as the rotor rotates and it allows larger time steps to be used [9]. The rotating region around the rotor had a wall boundary condition, so a turbulent model was used using K-Omega turbulence, to accurately capture behaviour near this 'wall' and regions of separation within it [10]. Standard boundary conditions were used and more details are found in [7].

4.2 Verification with mesh convergence

A mesh convergence study was conducted to find the appropriate mesh that accurately represents the solution of the problem. By increasing the base size on the 'fluid mesh', the results were not significantly affected, so this study focused on the 'rotating mesh' as this is the mesh that models the boundary layer and wake, which is significant. The study here shows values for torque and thrust when run for fixed RPM of 2500 and climb velocity of 1m/s. The minimum base size was set to 10% and base size was gradually reduced from 5.5m to 3m. Reducing the base size beyond this was restricted by computational power (the simulation would shut down).

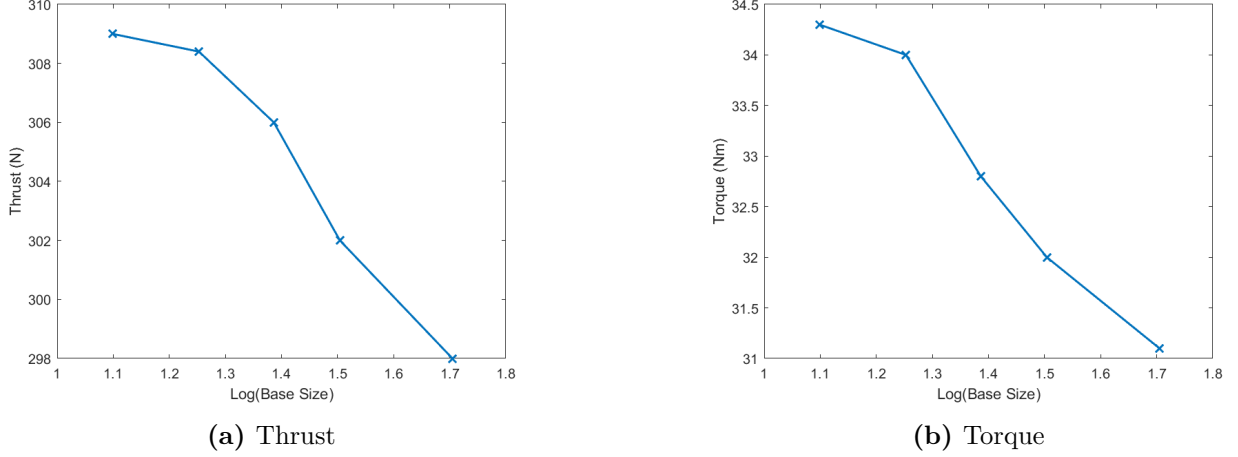


Figure 10. Final Geometry

From Figure 10, we see that values start to converge at a base size of 3.5m. Ideally further iterations would be run but this was not possible due to computational limits and licensing hour limits. Also the trend suggests that reducing base size below this only results in minimal improvements and does not justify the extra computational complexity involved. A 1000 fixed-step stop criteria was imposed to prevent excessively long runs.

4.3 Discussion of CFD Results and comparison to BEM

Table 10 compares the results in CFD and BEM predictions when compared for a fixed rpm of 2500 and climb velocity of 1m/s. There is a significant percentage difference in thrust predictions of 31.8%, however the difference in torque is only 10% - comparable to the validation results in [7] where the torque error is 8.8%. BEM is likely to overestimate torque since it assumes 2D data which is different from the 3D results. Hence, rotational augmentation caused by 3D turbulence effects is not accounted for and separation is not modelled. This implies that tangential blade forces are overpredicted and hence torque is overpredicted slightly [11]. The point of adding stall effects is further described in [12].

Table 10. Comparison of CFD and BEM

	Thrust (N)	Torque (Nm)
CFD	309	34.3
BEM	449	38.2

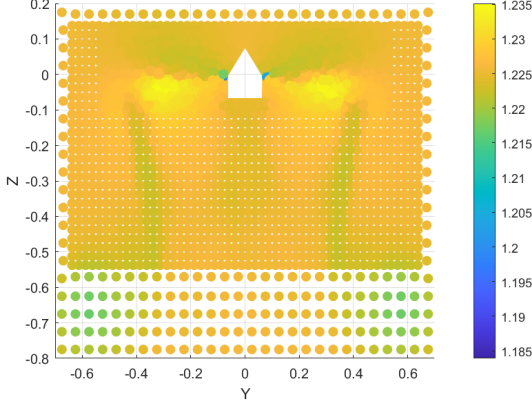
To ensure the rotors can still operate as required, the operating RPM of the drone can simply be shifted and this is discussed later. However the difference in thrust predictions is significant and causes should at least be investigated within the scope of the project.

Potential issues with the CFD setup include the model being based on a much smaller rotor and some parameters being set through trial and error [13]. Due to computational complexity and time available, extensive parameter experimentation wasn't feasible. However, accurate torque predictions indicate the CFD setup is likely correct and the discrepancies are more likely due to breakdowns in the assumptions and models used in BEM at high RPMs which we can now discuss. Due to limited space, the hub loss investigation was moved to the appendix.

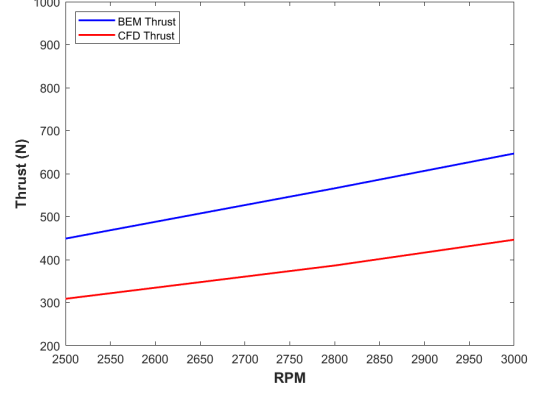
4.3.1 Investigating Compressibility and Viscosity effects

The inviscid assumption of BEM leads to the wake and wake rotation not being accurately modelled which is significant for lightly loaded rotors due to high induced velocities [12]. Also, flow separation cannot be modelled even though there are clear areas of recirculation and high adverse pressure

gradients seen in the Figure in Appendix D. Compressibility could potentially become significant since at an RPM of 2500, the tip mach number exceeds 0.3. Furthermore, BEM assumes there's no interactions between blade elements which lacks validity at high RPMs that the VTOL rotors operate at.



(a) Density Along Plane Section of Rotor CFD



(b) Thrust RPM Comparison

Figure 11. CFD data

To investigate compressibility effects in CFD, we can consider Figure 11a which showcases the density variation in the rotor plane. The legend highlights how the density in the wake of the rotor falls as much as 3.3%. These wake effects and density loss is not accounted for within BEM, but it does not justify such a large discrepancy in thrust predictions and is unlikely to be significant. To further investigate this, a STAR-CCM model was run with a constant density assumption with $\rho = 1.225 \text{ kg/m}^3$ but the change in thrust prediction was insignificant as expected (2% increase). Furthermore, we see how the end of the wake region falls into a coarser mesh so may not be predicted as accurately- in future simulations, the inner 'rotating' mesh should thereby be enlarged to incorporate the wake profile into it. A final simulation was run with both incompressible and inviscid models, however this did not make any significance to the results and a dynamic viscosity scene can be found in Appendix D.

4.3.2 Low RPM results

From the initial literature review, it was clear that BEM has been validated against many experiments, however most of the data is from wind turbine experiments, which are typically operated at significantly lower RPMs. Since the impacts of constant density and inviscid models cannot bring CFD and BEM results closer together, the next logical step was to run the simulation at a significantly lower RPM - the selected RPM was 1000. A comparison of these results to BEM is shown in Table 11. The error in thrust predictions is now only 4.1% which suggests that BEM theory is much better at predicting values at a lower RPM. With more time, more simulations would be run to at a large range of RPMs to see up to which point BEM can reasonably match CFD results.

Table 11. Comparison of CFD and BEM

	Thrust (N)	Torque (Nm)
CFD	67.1	6.5
BEM	70	6.1

4.3.3 New operating region

Figure 11b highlights differences in thrust predictions for the rotor at different RPMs between CFD and BEM. The average difference in thrust predictions is approximately 31% at the RPMS tested of 2500, 2800 and 3000, and the a best line has been fitted. Further RPM values were not tested since licencing hours ran out on STAR-CCM.

5 Conclusion

This report has achieved its aims by providing a blueprint for rotor design methodology with parameter choices being determined from a combination of both key parametric studies, mission requirements and requirements from other teams. The report then discusses the performance with both computational and analytical methods utilised and evaluated against each other.

Analytical methods were required for the rotor design, however they fell short in performance analysis for thrust estimates, perhaps due to faults within theoretical assumptions- within the timeline and computational limits, only some of these modelling assumptions could be investigated and were not found to be significant in matching the analytical and computational methods.

Section 4.3.2 suggests that analytical methods may not be suitable at the RPM that this design will operate at. However the report acknowledges that more data should be gathered and tested at RPMs < 1000 to validate this statement since it was expected that BEM to provide closer results. Further assumptions of BEM should also be tested to try and account for the thrust prediction discrepancies which are observed at higher RPMs.

Overall, the goal of the design was to produce a rotor capable of achieving a thrust of 412N/rotor within the power capacity available. Taking CFD as the most conservative measure of performance, it is now required that to produce the target thrust, the RPM must be increased to 2900. This would increase the power requirement to 12.7kW- this is very feasible with the current motor choice discussed in [14], although in future iterations this would be minimized.

6 Future Improvements

Ideally, experimental testing for a single rotor could be used to validate performance and also validate between CFD and BEM predictions, although the cost would be significant. In a second design iteration, the rotor design parameters could be adjusted, and a new hub design would be requested from the structures team. The rotor design timeline meant that structural issues in particular had to be anticipated. Since this design has proved to be structurally sound, the next blade design could be made thinner and the outboard airfoil design could be reselected purely for aerodynamic properties. In general, minimizing power requirements would be prioritized for the next design iteration. Furthermore, the transition region between the inboard and outboard aerofoils could be varied.

Future BEM performance predictions could add dynamic stall and wake models which are provided by [12]. Moreover, there were issues with importing the CAD onto Star CCM, as the geometry could not be modelled as one synchronous body so this issue should be fixed. Knowing now that BEM is likely to give overpredictions, the design thrust should be set higher, although increasing RPM of the current design is also sufficient since the Mach number at the tip would still be significantly below the transonic regime. With further time, more modelling assumptions could be investigated to fully understand the significant discrepancy between CFD and BEM at high RPMs. It would also be useful to attempt to model the rotor on Ansys Fluent to provide alternative CFD data, and as a confirmation that the rotor was modelled accurately using STAR CCM.

References

- [1] European Union Aviation Safety Agency. Air transport vtol capable aircraft, Jun 12 2024. URL <https://www.easa.europa.eu/en/domains/drones-air-mobility/drones-air-mobility-landscape/air-transport-vtol-capable-aircraft>.
- [2] HAL Open Science. The Blade Element Momentum theory (BEM) introduced by C.N.H. Technical report, HAL Open Science, 2020.
- [3] C. N. H. Adkins and R. H. Liebeck. Design of optimum propellers. Technical report, 1981.
- [4] Parasdeep Singh. Halo reach: Design and optimisation of pusher propeller. GDP report 02099678, Dept. of Aeronautical Engineering, Imperial College London, 2024.
- [5] M Sriti and Y. El khchine. Tip loss factor effects on aerodynamic performances of horizontal axis wind turbine, 2017. URL <https://www.sciencedirect.com/science/article/pii/S1876610217325547>. Accessed: 2024-06-12.
- [6] J. Gordon Leishman. *Principles of Helicopter Aerodynamics*. Cambridge University Press, 2000.
- [7] Matthew M Homa. Halo reach: Wing design and computational propeller analysis. GDP report 02038403, Dept. of Aeronautical Engineering, Imperial College London, 2024.
- [8] A new user’s guide to simcenter star-ccm+ simulation (part 3/5): Meshing, Jun 15 2024. URL <https://community.sw.siemens.com>.
- [9] A new user’s guide to simcenter star-ccm+ simulation, 2024. URL <https://community.sw.siemens.com>.
- [10] The k - omega SST Turbulence Model. URL <https://www.youtube.com/watch?v=myv-ityFnS4>.
- [11] A Bouhelal and A Smaili. Comparison of BEM and Full Navier Stokes CFD methods for Prediction of Aerodynamics Performance of HAWT Rotors. Technical report, École Nationale Polytechnique.
- [12] Moriarty P.J. and Hansen A.C. Aerodyn theory manual, 2005. URL <https://www.nrel.gov/docs/fy05osti/36881.pdf>. Accessed: 2024-06-14.
- [13] Austin R. Schenk. Computational Investigation of the Effects of Rotor-on-Rotor Interactions on Thrust and Noise . Technical report, Brigham Young University, 2020.
- [14] Tharm Chursuk. Halo reach: Optimising weight and balance through comprehensive powerplant sizing for vtol fixed-wing uav. GDP report 02048996, Dept. of Aeronautical Engineering, Imperial College London, 2024.
- [15] QBlade. Qblade documentation, 2024. URL <https://docs.qblade.org/>. Accessed: 2024-06-12.

Appendix

A BEM theory

To summarise:

- Actuator Disk theory assumes that the velocity through the rotor plane does not contain a tangential component and the pressures far upstream and downstream of the rotor are equal to the ambient pressure. Then through conservation of mass and momentum, the assumptions allow for the calculation of the rotor performance metrics[15].
- Blade Element Theory simply discretizes a blade radially into several spanwise sections (elements) and assumes the individual geometric and aerodynamic properties of each section (such as C_L , C_D and $(L/D)_{\max}$). The elements do not interact with one another [?].The blade forces can be expressed as a function of geometric (chord, radius) properties and axial and tangential induction factors so an iterative approach is used to solve for C_p and C_t , such that both individual theories are satisfied [15].

B Qblade/Xrotor/Thrust estimates

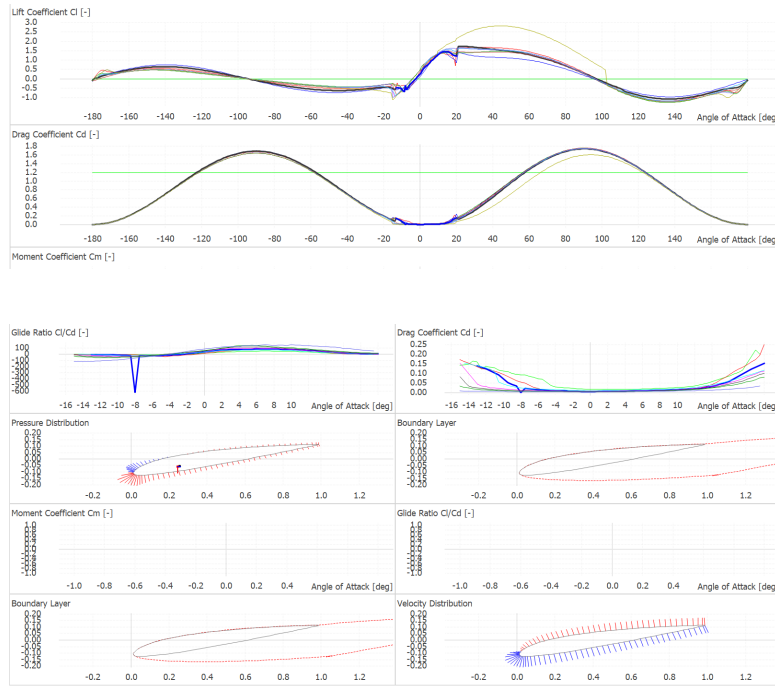
Thrust (N)
398

Table 12. Xrotor BEM performance

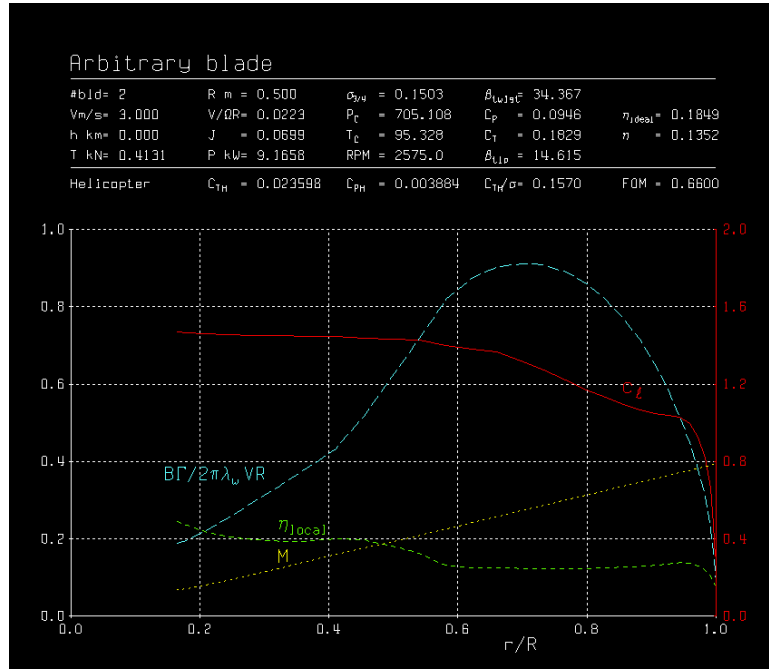
Thrust (N)
412

Table 13. In-house code prediction

QBlade interpolated airfoil data:



XROTOR data:



C Investigating Loss Model

It was initially thought that hub losses may be significant and this was not accounted for in the BEM model. In [12], a potential hub-wake loss model is proposed which is a correction for the induced velocity resulting from a vortex being shed near the hub of the rotor. The combined hub and tip loss factor can then be multiplied together. This was found to be insignificant.

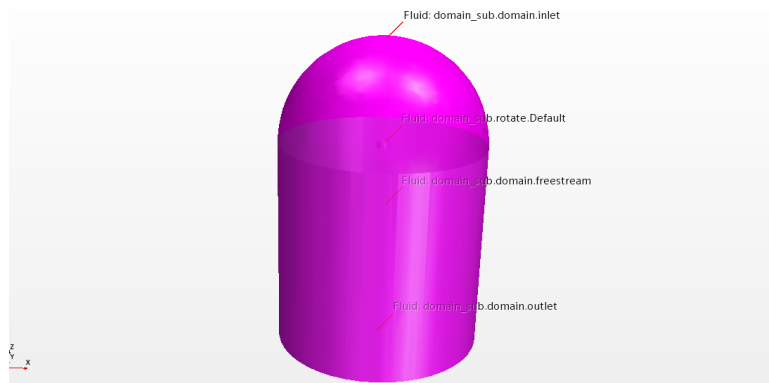
$$F_{hub} = \frac{2}{\pi} \arccos(e^{-f}) \quad (4)$$

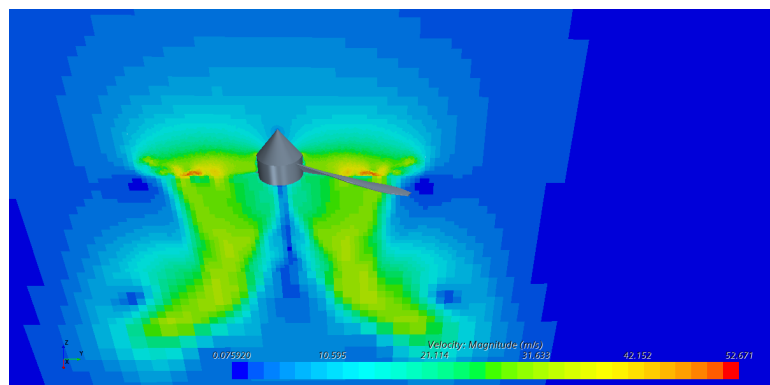
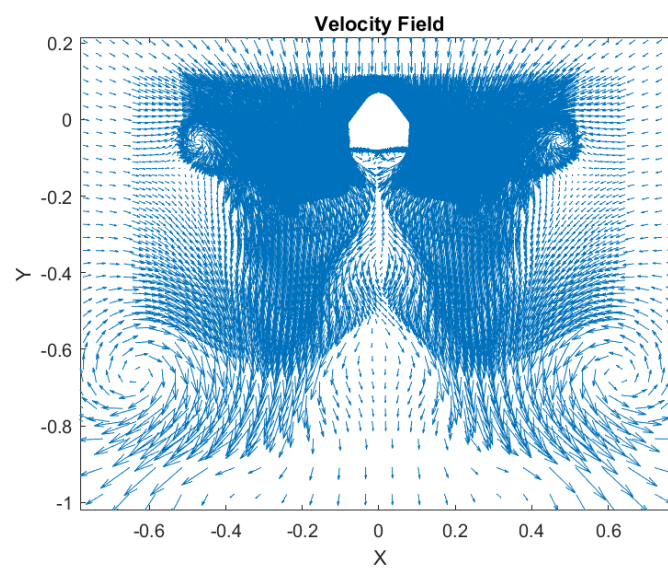
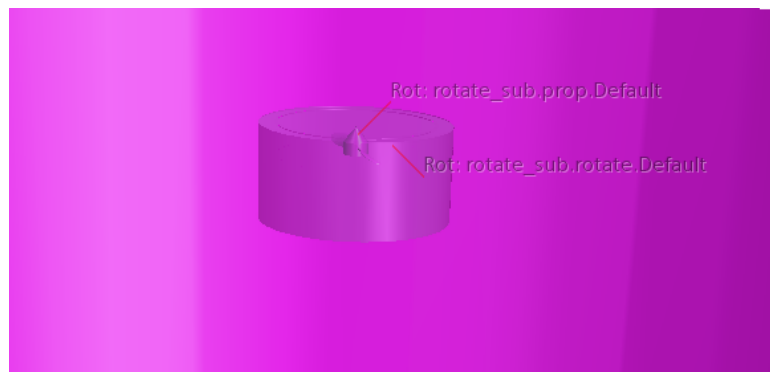
$$f_{hub} = \frac{B(r - R_{hub})}{2r \sin \varphi} \quad (5)$$

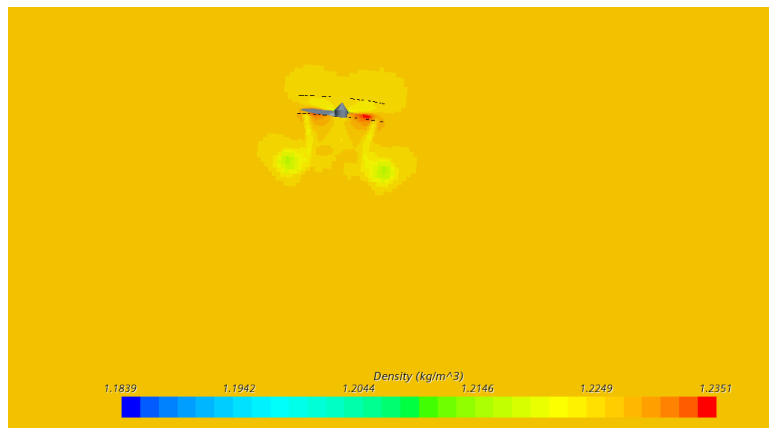
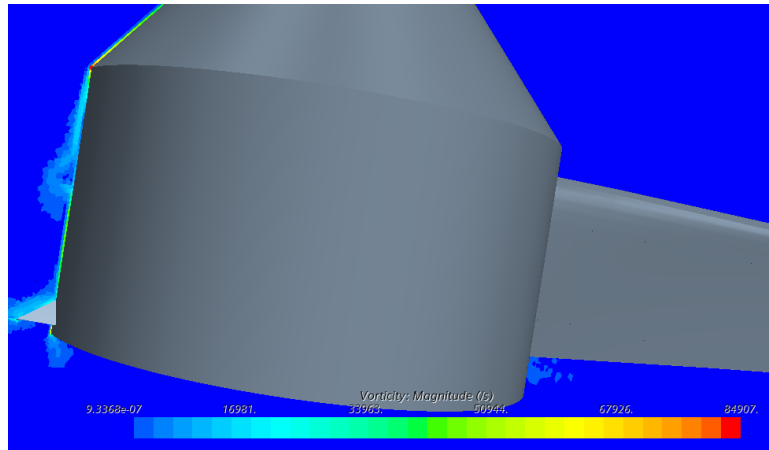
$$F = F_{hub} \cdot F_{tip} \quad (6)$$

D CFD

Pictures of geometry and some scenes are shown below.







E Analytical Code

Some preliminary calculations.

```
% Define parameters
CL = .8; % Average coefficient of lift using helicopter dynamics
D = 1; % Diameter of the blade
R = D / 2; % Radius of the blade
A = R * 0.1; % Blade area estimation (0.5 * chord * R)
Nb = 2; % Number of blades
W = 412.4 / Nb; % Lift by each rotor, total thrust/4 = 412.4
rho = 1.225; % Air density (kg/m^3)
chord = 0.1; % Chord length of the blade (m)
a = 343; % Speed of sound in air (m/s)
mu = 1.85e-5;

% Calculate velocity and required omega
V = sqrt((W * 2) / (rho * CL * A)); % Velocity at 2/3 along radius
omega_required = V / ((2/3) * R); % Required angular velocity (rad/s)
rpm = omega_required * 60 / (2 * pi) % Convert to RPM

% Blade design
R = linspace(0, D/2, 100); % Radial positions along the blade
omega = omega_required; % Angular velocity in rad/s
U = omega * R; % Tangential velocity
M = U / a; % Mach number
```



```

Re = (rho .* U .* chord) / mu; % Reynolds number

% Creating the grid for contour plots
[x, y] = meshgrid(linspace(-R(end), R(end), 100), linspace(-R(end), R(end), 100));
r = sqrt(x.^2 + y.^2); % Radial distance from the center

% Interpolate Mach number values over the grid
M_grid = interp1(R, M, r, 'linear', 'extrap');
% Interpolate Reynolds number values over the grid
Re_grid = interp1(R, Re, r, 'linear', 'extrap');

% Set values outside the blade radius to NaN
M_grid(r > R(end)) = NaN;
Re_grid(r > R(end)) = NaN;

% Plotting the Mach number as a contour plot
figure;
contourf(x/R(end), y/R(end), M_grid, 'LineColor', 'none');
colorbar;
caxis([0 max(M)]); % Adjust color axis to the range of Mach numbers
xlabel('x/R');
ylabel('y/R');
%title('Mach Number (hover)');
axis equal;
set(gca, 'Color', 'w'); % Set the background color to white

% Plotting the Reynolds number as a contour plot
figure;
contourf(x/R(end), y/R(end), Re_grid, 'LineColor', 'none');
colorbar;
caxis([0 max(Re)]); % Adjust color axis to the range of Reynolds numbers
xlabel('x/R');
ylabel('y/R');
%title('Reynolds Number (hover)');
axis equal;
set(gca, 'Color', 'w'); % Set the background color to white

%blade lift generation
%lift distribution allong wing
rho=1.225
%get Cl distribution
% Read the data from the file
CL = readmatrix('Cl_distribution.txt')

% Extract the relevant data (skipping the first two rows)
CL = CL(3:end, 2)

%get velcoity distribution
% Read the data from the file
V = readmatrix('V_distribution.txt')

% Extract the relevant data (skipping the first two rows)
Radius = V(3:end, 1)

```

```
V = V(3:end, 2)
```

```
%get chord distribution
% Read the data from the file
Chord = readmatrix('Chord_distribution.txt ')

% Extract the relevant data (skipping the first two rows)
Chord = Chord(3:end, 2)

L_distrib=0.5*rho.*V.^2.*CL.*Chord
plot(Radius,L_distrib)
ylabel('lift per unit span')
xlabel('span')
```

E.1 Adkins optimisation

Below is some code used to optimise the chord and twist of which was then further optimised using QBlade. Further details on methodology used within [4].

```
clc
clear all
close all
```

```
Hub_r = 0.08; % Hub radius %
R = 0.5; % Tip radius
B = 2; % Number of blades %
stations = 100; % Finite number of stations
RPM = 2500;
```

```
V = 110; %tip velcoity as defined in helicopter dynamics
C_L = 1.3; % Design lift coefficient
alpha = 2.5;
epsilon = 0.01;
option = 'T';
value = 100;
```

```
%FM = 1/((C_T^(3/2)) / (sqrt(2)*C_P))
```

```
function [CL, chord, twist, eta_p, output_value, alpha_rad, epsilon, beta, zeta] = Adk
```

```
% Define the given parameters
rho = 1.225; % Air density
Hub_r = Hub_r; % Hub radius %
R = R; % Tip radius %
RPM = RPM;
omega = RPM * 2 * pi / 60; % Rotational rate/omega %
V = V; % Freestream velocity
B = B; % Number of blades %
stations = stations; % Finite number of stations
Cl = C_L; % Design lift coefficient for each station %
```

```

%Clark - Y
% Re = [50000, 100000, 200000, 500000, 1000000]; % Re's for aerofoil
% C_L = [1.2442, 1.0873, 0.9005, 0.8090, 1.1090];
% aoa = [9.25, 6.75, 4.75, 3.75, 6.75]; % aoa in degrees
% L_D = [29.6, 53, 73.2, 98.7, 114.8];
% C_L_Max = [1.3370, 1.3694, 1.3965, 1.4329, 1.5240];
% stall = [11.25, 12.5, 12.5, 13.5, 14.75];

% NACA-23012
% Re = [50000, 100000, 200000, 500000, 1000000];
% C_L = [1.4379, 1.3732, 1.2878, 1.1238, 1.0304];
% aoa = [8.75, 7, 5.5, 3.5, 2.5];
% L_D = [39.7, 66.5, 98, 143, 175.2];
% C_L_Max = [1.6133, 1.6491, 1.6277, 1.7038, 1.8311];
% stall = [12, 14.5, 16.75, 15.5, 17.5];
%

% E-216
Re = [50000, 100000, 200000, 500000, 1000000];
C_L = [1.4618, 1.3516, 1.1901, 1.1344, 1.0159];
aoa = [8, 6, 4, 3.25, 1.75];
L_D = [40.7, 68.5, 99.2, 136.8, 155.1];
C_L_Max = [1.56, 1.5165, 1.5575, 1.6634, 1.7600];
stall = [10.25, 10.0, 10.25, 12.25, 13.25];
%

epsilon = 1./L_D;

% Fit polynomials (using a second-order polynomial as an example)
p_CL = polyfit(Re, C_L, 2);
p_aoa = polyfit(Re, aoa, 2);
p_LD = polyfit(Re, L_D, 2);
p_epsilon = polyfit(Re, epsilon, 2);

% Requirements %
if option == 'T'
    T = value; % Thrust
    T_c = 2 * T / (rho * (V^2) * pi * (R^2)); % Thrust coefficient
elseif option == 'P'
    P = value * 0.8; % Power into the propeller
    P_c = 2 * P / (rho * (V^3) * pi * (R^2)); % Power coefficient
end

% Define other constants
% Step 1: Initial guess for zeta
zeta = 0; % Initial guess for zeta
tolerance = 0.001; % 0.1% tolerance
max_iterations = 1000; % Maximum number of iterations to prevent infinite loop
iteration = 0; % Initialize iteration counter

```

```

% Initialize variables
zeta_new = 0.1; % Ensure the loop starts
while abs(zeta_new - zeta) / zeta_new > tolerance && iteration < max_iteration
    zeta = zeta_new;

% Step 2: Calculate phi and F for each radial station
r = linspace(Hub_r, R, stations);

xi = zeros(1, stations);
phi = zeros(1, stations);
f = zeros(1, stations);
F = zeros(1, stations);
G = zeros(1, stations);
Wt = zeros(1, stations);
Wc = zeros(1, stations);
a = zeros(1, stations);
a_prime = zeros(1, stations);
W = zeros(1, stations);
chord = zeros(1, stations);
beta = zeros(1, stations);

for i = 1:stations
    xi(i) = r(i) / R; % Non-dimensional radius = lambda * x
end

lambda = V / (omega * R); % Speed ratio V/omega*R

phi_t = atan(V / (omega * R) * (1 + zeta / 2)); % Flow angle at the tip

for i = 1:stations
    phi(i) = atan((tan(phi_t)) / xi(i)); % Flow angle at each station
end

for i = 1:stations
    f(i) = (B / 2) * (1 - xi(i)) / sin(phi_t);
    F(i) = (2 / pi) * acos(exp(-f(i))); % Prandtl's tip loss factor
end
F(end) = 0;

% Step 3: Determine the product Wc
for i = 1:stations
    Wt(i) = V * zeta * sin(phi(i)) * cos(phi(i)); % Tangential velocity
    G(i) = F(i) * cos(phi(i)) * sin(phi(i)); % Circulation function
    Gamma = 2 * pi * (V ^ 2) * G(i) / (B * omega); % Circulation
    x = omega * r(i) / V; % Speed ratio

    Wc(i) = 4 * pi * lambda * V * R * zeta * G(i) / (Cl * B);
    kinematic_viscosity = 1.48 * 10^(-5);
    Reynolds_number = Wc(i) / kinematic_viscosity;
    CL(i) = 7;

    while (abs(CL(i) - Cl) > 0.01)

```

```

        Cl = CL(i);
        Wc(i) = 4 * pi * lambda * V * R * zeta * G(i) / (CL(i) * B);
        Reynolds_number = Wc(i) / kinematic_viscosity;
        CL(i) = polyval(p_CL, Reynolds_number);
    end

    % Step 4: Determine epsilon and alpha_design
    alpha_rad(i) = polyval(p_aoa, Reynolds_number) * pi / 180; % Angle of
    epsilon(i) = polyval(p_epsilon, Reynolds_number); % Drag-to-lift ratio

    % Step 6: Determine a, a' and W
    a(i) = (zeta / 2) * cos(phi(i))^2 * (1 - epsilon(i) * tan(phi(i)));
    % Axial interference factor
    a_prime(i) = (zeta / (2 * x)) * cos(phi(i)) * sin(phi(i)) * (1 + epsilon(i));
    % Rotational interference factor
    W(i) = V * (1 + a(i)) / sin(phi(i)); % Local total velocity

    % Step 7: Compute the chord and the blade twist
    chord(i) = Wc(i) / W(i); % Chord calculation based on Wc
    beta(i) = alpha_rad(i) + phi(i); % Twist distribution
end

% Step 8: Calculate derivatives
I_prime_1 = zeros(1, stations);
I_prime_2 = zeros(1, stations);
J_prime_1 = zeros(1, stations);
J_prime_2 = zeros(1, stations);

for i = 1:stations
    I_prime_1(i) = 4 * xi(i) * G(i) * (1 - epsilon(i) * tan(phi(i)));
    I_prime_2(i) = lambda * (I_prime_1(i) / (2 * xi(i))) * (1 + epsilon(i));
    J_prime_1(i) = 4 * xi(i) * G(i) * (1 + epsilon(i) / tan(phi(i)));
    J_prime_2(i) = (J_prime_1(i) / 2) * (1 - epsilon(i) * tan(phi(i))) * c
end

% Integrate the derivatives over xi using the trapezoidal rule
I_1 = trapz(xi, I_prime_1);
I_2 = trapz(xi, I_prime_2);
J_1 = trapz(xi, J_prime_1);
J_2 = trapz(xi, J_prime_2);

% Step 9: Determine zeta_new based on provided input
if exist('T_c', 'var')
    % Use the first set of equations if T_c is provided
    zeta_new = (I_1 / (2 * I_2)) - sqrt((I_1 / (2 * I_2))^2 - T_c / I_2);
    P_c = J_1 * zeta_new + J_2 * zeta_new^2;
elseif exist('P_c', 'var')
    % Use the second set of equations if P_c is provided
    zeta_new = -(J_1 / (2 * J_2)) + sqrt((J_1 / (2 * J_2))^2 + P_c / J_2);
    T_c = I_1 * zeta_new - I_2 * zeta_new^2;
else
    error('Please provide either T_c or P_c as an input. ');
end

```

```

        end

        % Increment iteration counter
        iteration = iteration + 1;
    end

    % Determine propeller efficiency as  $T_c / P/c$ 
    eta_p = T_c / P_c;

    % Return the chord and twist distribution
    twist = beta;

    if option == 'T'
        %  $P = P_c * (\rho * ((RPM / 60) ^ 3) * (R * 2) ^ 5);$ 
        P = P_c * (rho * (V^3) * pi * (R^2)) / 2;
        output_value = P;
    elseif option == 'P'
        %  $T = T_c * (\rho * ((RPM / 60) ^ 2) * (R * 2) ^ 4);$ 
        T = T_c * (rho * (V^2) * pi * (R^2)) / 2;
        output_value = T;
    end
end
end

```